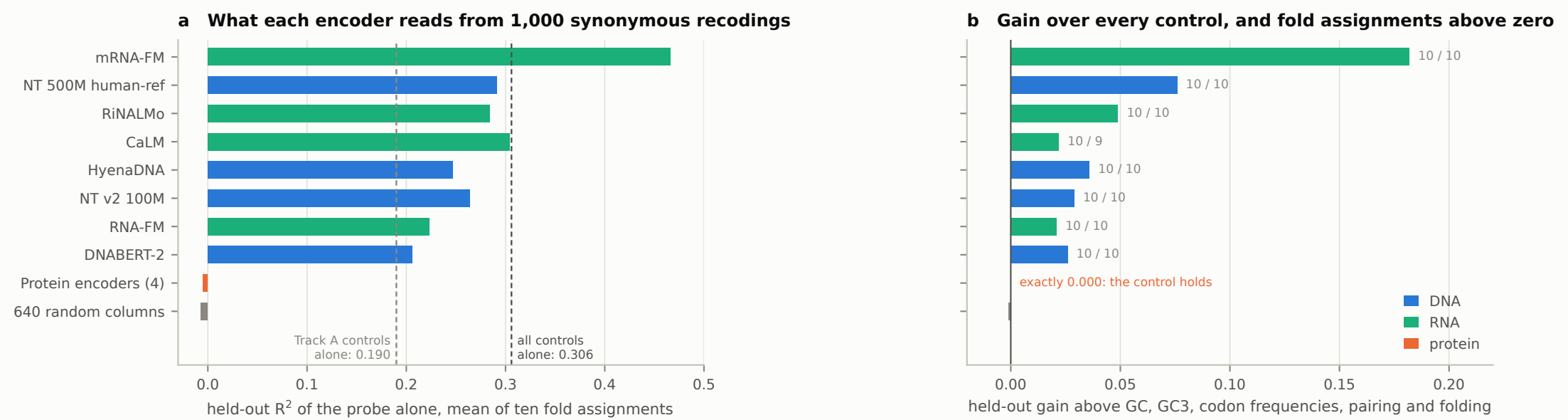
a What each encoder reads from 1,000 synonymous recodings
mRNA-FM
NT 500M human-ref
RiNALMo
CaLM
HyenaDNA
NT v2 100M
RNA-FM
DNABERT-2
Protein encoders (4)
640 random columns
Track A controls alone: 0.190
all controls alone: 0.306
0.0
0.1
0.2
0.3
0.4
0.5
held-out R2 of the probe alone, mean of ten fold assignments
b Gain over every control, and fold assignments above zero
10 / 10
10 / 10
10 / 10
10 / 9
10 / 10
10 / 10
10 / 10
10 / 10
exactly 0.000: the control holds
DNA
RNA
protein
0.00
0.05
0.10
0.15
0.20
held-out gain above GC, GC3, codon frequencies, pairing and folding